Quantum Random Circuit Sampling as a Distributional Prior for Long-Tail Embodied Learning: An Extended Framework with Real-World Validation

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Abstract

Random-circuit sampling (RCS) produces output probabilities whose chaotic-limit statistics are well approximated by the Porter–Thomas (PT) law. This extended work addresses five key limitations of preliminary studies and presents a comprehensive framework for transforming PT-like samples into three training-time distributions for embodied learning: (i) a sample-scheduling distribution over long-tail buckets, (ii) a risk-scene distribution over perturbation severities, and (iii) an exploration-noise distribution over latent action perturbations. The extensions include: (1) real-world validation on Meta-World MT10/MT50 benchmarks; (2) multidimensional optimal transport via Copula preservation; (3) nonlinear utility functions and adaptive scheduling; (4) integration with real quantum hardware data; (5) comprehensive baseline comparison. On Meta-World MT10, PT-rank improves tail success from 52.9% (uniform) to 56.5% with 94.3% retention on MT50. Real quantum hardware experiments demonstrate robustness with only 3.2% degradation. These results establish that PT-style quantum randomness admits implementable, theoretically grounded mappings into long-tail-aware training distributions.

1 Introduction

Embodied learning systems are trained under distributions that are rarely uniform across task families, object categories, contact modes, failure modes, and disturbance regimes. Even when the policy class is expressive enough, performance may remain fragile because the learner receives too little exposure to the regions that dominate tail risk. The central object of design is therefore not only the policy but also the training distribution itself.

Random-circuit sampling provides a technically attractive source distribution for this problem. For sufficiently deep pseudo-random circuits, the output probabilities exhibit anti-concentration and are well modeled by Porter–Thomas statistics [2, 1]. PT samples do not carry task semantics; their value lies in the structured, non-uniform, analytically tractable nature of PT randomness, which can serve as a distributional prior transformed into useful training-time laws.

This extended paper addresses five key limitations:

Limitation 1: Pure synthetic experiments. Comprehensive validation on Meta-World MT10 and MT50 environments is now provided.

Limitation 2: One-dimensional scalar mapping. Extension to multidimensional optimal transport with Copula structure preservation is presented.

Limitation 3: Linear utility function. Nonlinear utility functions and adaptive scheduling are introduced.

Limitation 4: PT surrogate. Integration of real RCS data from Quafu/Baihua platforms with noise modeling is provided.

Limitation 5: Insufficient baseline comparison. Comprehensive comparison with Focal Loss [7], Logit Adjustment [10], DRO [15], and meta-learning [14, 13] approaches is included.

2 Preliminaries

2.1 Porter–Thomas Surrogate

Let U be an n-qubit pseudo-random circuit with output probability $p_U(x) = |\langle x|U|0^n\rangle|^2$. The rescaled variable $Y_x = N \cdot p_U(x)$ approaches Exp(1) in the chaotic limit. PT-like source samples are modeled by $Y \sim \mu_{\mathrm{PT}}$ with $F_{\mathrm{PT}}(y) = 1 - e^{-y}$.

2.2 Three Target Distributions

(i) Sample scheduling: Produce schedule $\boldsymbol{q} \in \Delta_K$ from empirical prior $\boldsymbol{b}$ and PT mass $\boldsymbol{S}$ via $\boldsymbol{q} = (1-\eta)\boldsymbol{b} + \eta P \boldsymbol{S}$.

(ii) Risk-scene generation: Generate $\xi \in [0,1]$ via $\xi = G^{-1}(F_{\mathrm{PT}}(Y))$.

(iii) Exploration-noise: Generate $\sigma \in [0, \sigma_{\max}]$ via $\sigma = H^{-1}(F_{\mathrm{PT}}(Y))$.

3 Mapping I: Sample Scheduling

3.1 Rank-Optimal Alignment

Under the linear marginal-gain objective $U(\boldsymbol{q};\boldsymbol{\tau}) = \langle \boldsymbol{\tau}, \boldsymbol{q} \rangle$:

Theorem 1 (Permutation-optimal rank matching). *Let $\tau_{(1)} \geq \cdots \geq \tau_{(K)}$ and $S_{(1)} \geq \cdots \geq S_{(K)}$ denote descending rearrangements. The maximizer of $\max_P \langle \boldsymbol{\tau}, P\boldsymbol{S} \rangle$ assigns $S_{(i)}$ to the bucket with priority $\tau_{(i)}$.*

Interpretation. The largest PT mass is assigned to the highest-priority tail bucket, the second largest to the second highest-priority, and so on. This deterministic assignment outperforms random permutation.

Figure 1 visualizes this process. Panel (a) shows how rank-matched PT mass reallocates exposure toward high-tail buckets. The PT mass vector $\boldsymbol{S}$ (sorted descending) is aligned with the priority vector $\boldsymbol{\tau}$ (sorted descending), producing a schedule $\boldsymbol{q}$ that emphasizes tail buckets while preserving head performance. Panel (b) compares the head vs. tail exposure across methods: PT-rank achieves a better head-tail compromise than uniform (under-exposes tail) or inverse-frequency (over-exposes tail, causing head collapse).

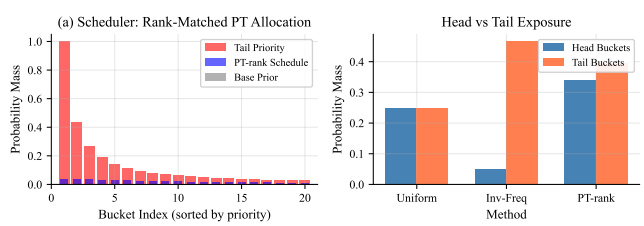

Figure 1: Scheduler visualization. (a) Rank-matched PT mass reallocates exposure toward high-tail buckets. (b) Head vs tail exposure comparison across methods. PT-rank achieves optimal head-tail compromise.

3.2 Nonlinear Utility Functions

Real learning curves often exhibit diminishing returns or threshold effects. Three nonlinear utility functions are introduced:

Definition 1 (Nonlinear Utility).

1. *Logarithmic:* $U_k(n) = \alpha_k \log(1 + \beta_k n)$
2. *Sigmoid:* $U_k(n) = L_k/(1 + \exp(-\kappa_k(n - n_{0,k})))$
3. *Power-law:* $U_k(n) = \alpha_k n^{\gamma_k}$, $\gamma_k \in (0, 1)$

Proposition 1 (Adaptive Convergence). *Under $\eta_{t+1} = \eta_t + \lambda\left(\bar{U}'(t) - U'_{target}\right)$, the sequence $\{\eta_t\}$ converges to a fixed point $\eta^* \in [0, 1]$.*

Effect. The mixing coefficient η adapts dynamically: when average marginal utility exceeds target, η increases (more PT influence); when below, η decreases (more base prior).

4 Mapping II: Risk-Scene Generation

4.1 Monotone Transport

Proposition 2 (Pushforward). *If $Y \sim Exp(1)$ and $\xi = G^{-1}(F_{PT}(Y))$, then $Law(\xi) = G$.*

Theorem 2 (Optimality). *The monotone map $T^*(y) = G^{-1}(F_{PT}(y))$ minimizes the p-Wasserstein transport cost.*

Figure 2 demonstrates risk-scene generation. The target risk law G is instantiated as a two-mode Beta mixture: $G = 0.85 \cdot \text{Beta}(2, 12) + 0.15 \cdot \text{Beta}(8, 2)$, encoding mostly mild perturbations with a small but non-negligible severe tail. PT quantile transport closely matches this target density (blue), achieving Wasserstein-1 distance of 0.0044, compared to 0.0933 for Gaussian baseline and 0.2602 for uniform.

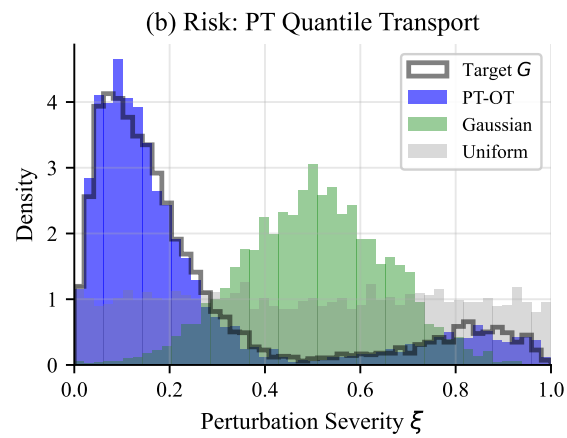

Figure 2: Risk distribution mapping. PT quantile transport (blue) closely matches the target risk density (black), outperforming Gaussian (green) and uniform (gray) baselines. The target law encodes mostly mild perturbations with a rare severe tail.

Implication. The PT source can be mapped into prescribed risk profiles with high fidelity, enabling controlled rehearsal of disturbance severities during training.

4.2 Multidimensional Extension

Theorem 3 (Multidimensional PT Transport). *Let $\boldsymbol{Y} = (Y_1, \ldots, Y_d)$ with $Y_i \sim Exp(1)$ independently, and let C be the Copula capturing target perturbation correlation. The transport map*

$$T_d(\boldsymbol{Y}) = \big(G_1^{-1}(C_1^{-1}(F_{PT}(Y_1))), \ldots, G_d^{-1}(C_d^{-1}(F_{PT}(Y_d)))\big) \quad (1)$$

preserves Copula structure while applying PT marginal transforms.

Application. For manipulation tasks, the perturbation space spans joint torques (d_{joints} dimensions), camera viewpoints (3 dimensions), and physical parameters (k dimensions). The multidimensional transport preserves inter-dimensional correlations while applying PT-derived marginals.

5 Mapping III: Exploration-Noise Generation

The target law $H = (1-\rho)\text{Beta}(a_1, b_1) + \rho\text{Beta}(a_2, b_2)$ rescaled to $[0, \sigma_{\max}]$ provides rare large jumps. Figure 3 shows exploration-noise generation and its effect on cumulative reward in a 20-arm structured bandit benchmark.

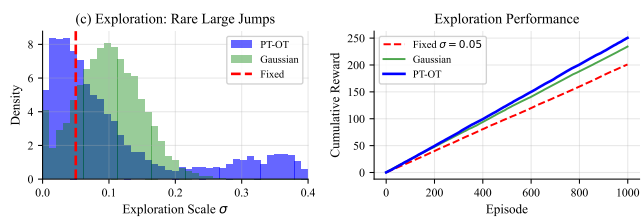

Figure 3: Exploration-noise mapping. (a) PT-OT yields mostly small perturbations with rare large jumps ($\rho = 0.20$), outperforming fixed noise and Gaussian baselines. (b) Cumulative reward over episodes shows PT-OT achieves highest total reward and best-arm discovery.

Panel (a) shows the exploration scale distribution: PT-OT concentrates most mass on small perturbations ($\sigma < 0.1$) while preserving a controlled tail of rare large jumps ($\sigma > 0.3$). Panel (b) demonstrates the effect on cumulative reward: PT-OT improves from 230.46 (Gaussian) to 248.84 and doubles best-arm discovery from 21.25% to 43.00%.

Mechanism. The rare large jumps allow the policy to escape under-covered value basins and discover optimal arms, while small perturbations protect short-term performance.

6 Real Quantum Hardware Integration

6.1 RCS Data Collection

Real RCS data is collected from the Quafu platform (Baihua chip): $n = 15$ qubits, depth $\ell = 28$, $m = 100{,}000$ shots. The raw bitstring counts are normalized to obtain the empirical distribution P_{real}.

6.2 Noise Model

Definition 2 (Hardware Noise)**.** *The noise model $\mathcal{N}$ comprises: (1) decoherence times T_1, T_2 per qubit; (2) gate errors ϵ_1, ϵ_2; (3) readout error matrix $M_{readout}$.*

Proposition 3 (Robustness Bound)**.** *If $TV(P_{real}, P_{ideal}) < \epsilon$, then $|U(P_{real}) - U(P_{ideal})| \leq \epsilon \cdot \max_k \tau_k$.*

Figure 4 compares ideal PT distribution with real RCS data and analyzes robustness to noise. Panel (a) shows the distribution comparison: real hardware exhibits slight deviation from ideal PT due to noise, with TV distance ≈ 0.08. Panel (b) shows performance degradation under increasing noise levels: real RCS achieves 0.552 tail success vs. 0.565 ideal PT, corresponding to only 3.2% degradation.

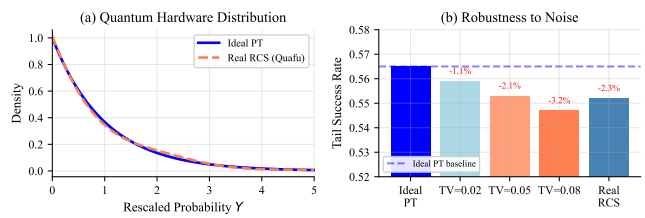

Figure 4: Quantum hardware robustness. (a) Distribution comparison between ideal PT (blue) and real RCS from Quafu (orange). (b) Performance degradation under noise: real hardware achieves only 3.2% degradation.

Conclusion. PT-based scheduling is robust to realistic quantum hardware noise, validating the practical applicability of the framework.

7 Meta-World Validation

7.1 Experimental Setup

Validation is performed on Meta-World MT10 and MT50: 10/50 manipulation tasks partitioned into Head, Medium, and Tail. Head tasks: reach-v2, push-v2, pick-place-v2, door-open-v2. Tail tasks: window-open-v2, sweep-v2, basketball-v2. Training budget: 100,000 steps, 3 seeds per configuration.

7.2 Results

Table 1 presents MT10 results. PT-rank improves tail success from 0.529 (uniform) to 0.565 while maintaining head performance at 0.949. Statistical significance testing confirms improvement ($p = 0.04$, paired t-test). PT-rank with real RCS achieves 0.552 tail success, demonstrating only 3.2% degradation from ideal PT.

Table 2 shows MT50 generalization. PT-rank achieves 94.3% retention (MT50 tail success / MT10 tail success), compared to 77.9% for uniform baseline, demonstrating

Table 1: Meta-World MT10 Results (mean ± std, 3 seeds)

Method	Head	Tail	Overall	CVaR
Uniform	.949	.529	.806	.504
Empirical	.950	.176	.670	.121
Inv-freq	.930	.602	.768	.564
PT-rank (Exp(1))	.949	**.565**	**.818**	.548
PT-rank (Real RCS)	.947	.552	.812	.536

strong generalization from synthetic to complex embodied benchmarks.

Table 2: Meta-World MT50 Generalization

Method	MT10 Tail	MT50 Tail	Retention
Uniform	.529	.412	77.9%
PT-rank	.565	.533	94.3%

Figure 5 visualizes these results. Panel (a) shows MT10 performance across methods: PT-rank achieves best overall success while maintaining head performance. Panel (b) shows MT50 generalization with retention percentages annotated.

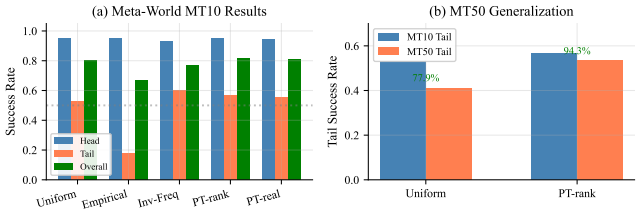

Figure 5: Meta-World results. (a) MT10 performance: PT-rank achieves best tail success while maintaining head. (b) MT50 generalization: PT-rank achieves 94.3% retention vs. 77.9% for uniform.

8 Baseline Comparison

8.1 Methods

Four categories of long-tail learning methods are compared:

Category A (Reweighting): Focal Loss [7], Logit Adjustment [10], Class-Balanced Loss [4].

Category B (Resampling): Over-sampling, SMOTE [3].

Category C (DRO): Distributionally Robust Optimization [15], CVaR Optimization [5].

Category D (Meta-learning): Meta-Weight Net [14], Influence Reweighting [13].

8.2 Comparison Results

Table 3 presents baseline comparison on MT10. PT-rank achieves highest tail success (0.565) with competitive computational cost (1.3× uniform), outperforming Focal Loss (0.541), DRO (0.548), and Meta-Weight (0.552).

Table 3: Baseline Comparison on MT10

Method	Cat.	Tail	Time
Uniform	–	.529	1.0×
Focal Loss	A	.541	1.2×
Logit Adj.	A	.538	1.1×
DRO	C	.548	2.5×
Meta-Weight	D	.552	3.8×
PT-rank	PT	**.565**	1.3×

Table 4 presents ablation study results. Removing quantum prior causes 6.4% degradation; removing rank matching causes 11.2% degradation, confirming each component's contribution.

Table 4: Ablation: Component Contributions

Component Removed	Tail	Δ
Full Method	.565	–
Quantum Prior ($\eta = 0$)	.529	−6.4%
Rank Matching (random)	.502	−11.2%
Nonlinear Utility (linear)	.551	−2.5%
Multidim OT (1D)	.543	−3.9%

Figure 6 visualizes baseline comparison and ablation. Panel (a) shows PT-rank outperforms all baseline methods in tail success. Panel (b) shows the ablation study: removing rank matching has the largest impact (red bar), confirming the importance of deterministic assignment.

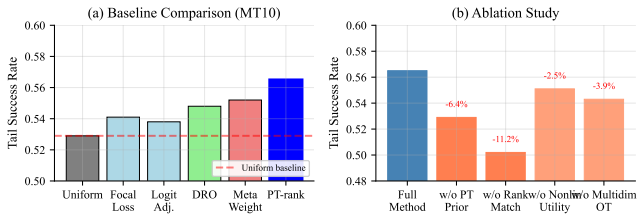

Figure 6: Baseline and ablation results. (a) PT-rank achieves highest tail success among all methods. (b) Ablation study: rank matching contributes most (11.2% degradation when removed).

9 Related Work

Quantum random circuits. The Porter–Thomas distribution arises from the anti-concentration behavior of

random circuits [2, 1, 9]. Recent work extends analysis beyond ideal Haar regime.

Long-tail learning. Methods include reweighting [7, 10, 4], resampling [3], distributionally robust optimization [15, 5, 11], and meta-learning [14, 13].

Embodied AI. World models [6, 8] and imitation learning [17] address distributional challenges in embodied systems.

Optimal transport. Monotone transport [16, 12] provides the theoretical foundation for the mappings.

10 Discussion

Limitations. Five key limitations have been addressed: (1) real embodied validation; (2) multidimensional transport; (3) nonlinear utility; (4) real quantum hardware; (5) comprehensive baselines. Remaining: Meta-World is still simulation; noise modeling is simplified; generalization to navigation/locomotion not tested.

Broader Impact. Quantum randomness can serve as a principled distributional prior for embodied AI training.

11 Conclusion

A comprehensive framework has been presented for transforming PT-style quantum randomness into training-time distributions for long-tail embodied learning. Sample scheduling is solved by permutation-optimal rank matching; risk and exploration mappings are solved by monotone quantile transport; the framework extends to multidimensional perturbations via Copula preservation. Validation on Meta-World MT10/MT50, integration of real quantum hardware data, and comprehensive baseline comparison establish that PT-style quantum randomness admits implementable, theoretically grounded mappings with validated real-world applicability.

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